

Foundations

James P. Ochoa

Theorem Sequence

We begin with some formal rules of logic. A statement is a sentence which, in a given context, is either true or false.

Logic Rule 1: No unstated assumption may be used in a proof.

Logic Rule 2: Let p and q be statements. If the compound statement “ p and not q ” implies a contradiction, then the statement “If p , then q ” is true.

We may use Logic Rule 2 to prove the statement in the following example.

Example 3: Let n be a positive integer. If 9 divides n , then 3 divides n .

Proof: Suppose that 9 divides n and 3 does not divide n . Since 3 does not divide n , neither does 3^2 . Since $3^2 = 9$, it follows that 9 cannot divide n . This is a contradiction, since 9 divides n . Therefore, if 9 divides n , then 3 divides n .

Logic Rule 4: Let p be a statement. The negation of the statement “Not p ” means the same as p .

Logic Rule 5: Let p and q be statements. The negation of the statement “If p , then q ” means the same as “ p and not q ”.

Logic Rule 6: Let p and q be statements. The negation of the statement “ p and q ” means the same as “Not p or not q ”. The negation of the statement “ p or q ” means the same as “Not p and not q ”.

Let $p(x)$ be a sentence about x (in this case, x is called a variable). We say that z is in the domain of p if the sentence $p(z)$ is a statement. For example, let $p(x)$ be the sentence “ $x + 3 = 7$ ”. Then 6 is in the domain of p , since $p(6)$ is the (false) statement “ $6 + 3 = 7$ ”. On the other hand, w is not in the domain of p , since we cannot in the current context determine whether “ $w + 3 = 7$ ” is true or false.

Let $p(x)$ be a sentence about x . The sentence “For all x , $p(x)$ ” is a statement. The expression “for all” is called the universal quantifier. It is understood that the phrase “for all x ” refers only to x in the domain of $p(x)$.

Logic Rule 7: Let $p(x)$ be a sentence about x . The negation of the statement “For all x , $p(x)$ ” means the same as “There exists an x such that not $p(x)$ ”.

Logic Rule 8: Let $p(x)$ be a sentence about x . The negation of the statement “There exists an x such that $p(x)$ ” means the same as “For all x , not $p(x)$ ”.

Logic Rule 9: Let p and q be statements. Suppose the statements “If p , then q ” and p are steps in a proof. Then q is a valid step.

Logic Rule 10: Let p , q , and r be statements. The following are all true statements:

- (1) If p , then p or q .
If q , then p or q .
- (2) If p and q , then p .
If p and q , then q .
- (3) If “Not q ” implies “Not p ”, then p implies q .
If p implies q , then “Not q ” implies “Not p ”.
- (4) If p implies q and q implies r , then p implies r .

Logic Rule 11: For every statement p , either p or “Not p ” is true.

Logic Rule 12: Let p , q , and r be statements. Let 1 denote a true statement, and let 0 denote a false statement. Then:

- (1) The statement “ p and q ” means the same as “ q and p ”,
- (2) The statement “ p or q ” means the same as “ q or p ”,
- (3) The statement “ p and q ; and r ” means the same as “ p ; and q and r ”,
- (4) The statement “ p or q ; or r ” means the same as “ p ; or q or r ”,
- (5) The statement “ p and q ; or r ” means the same as “ p or r ; and q or r ”,
- (6) The statement “ p or q ; and r ” means the same as “ p and r ; or q and r ”,
- (7) The statement “ p or 0” means the same as “ p ”,
- (8) The statement “ p and 0” is false,
- (9) The statement “ p or 1” is true,
- (10) The statement “ p and 1” means the same as “ p ”,
- (11) The statement “ p or not p ” is true, and
- (12) The statement “ p and not p ” is false.

The following will remain undefined: *set*, *is a member of*, *collection*, *a sentence about*, *statement*, *natural number*, and *integer*.

Let A be a set. If x is a member of A , we write $x \in A$. If x is not a member of A , we write $x \notin A$.

Definition 13: Let A and B be sets. We say that A is a *subset* of B , and write $A \subseteq B$, if the following statement is true:

If $x \in A$, then $x \in B$.

For example, let $A = \{1, 3, 5\}$ and let $B = \{1, 2, 3, 4, 5\}$. Then $A \subseteq B$.

Let S be a set, and let $p(x)$ be a sentence about members x of S . We write $\{x \mid p(x)\}$ to denote the collection of all $x \in S$ such that $p(x)$ is a true statement. In this case, $p(x)$ is called a *condition*.

Axiom 14: Let S be a set, and let $p(x)$ be a sentence about members x of S . Then $\{x \mid p(x)\}$ is a subset of S .

Definition 15: Let A and B be sets. We say that A and B are equal, and write $A = B$, if $A \subseteq B$ and $B \subseteq A$.

For example, let $A = \{2, 4, 6, 8\}$ and let $B = \{8, 6, 4, 2\}$. Then A and B are equal.

We write $\mathbf{N}$ to denote the set of natural numbers.

We write $\mathbf{Z}$ to denote the set of integers.

Axiom 16: The set of natural numbers is a subset of the set of integers.

The empty set $(\emptyset)$ is the set with no members.

Proposition 17: Let S be a set. Then $\emptyset \subseteq S$, and $S \subseteq S$.

Definition 18: Let S be a set. The power set of S , $\mathcal{P}(S)$, is the collection of all subsets of S . In this context, the set S is called the *universal set*.

Axiom 19: Let S be a set. Then $\mathcal{P}(S)$ is a set.

Problem 20: Let $S = \{2, 3, 4\}$. List the members of $\mathcal{P}(S)$.

Definition 21: Let S be a set. Let A and B be subsets of S .

(1) The intersection of A and B is the set

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

(2) The union of A and B is the set

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

(3) The complement of A is the set

$$\complement A = \{x \mid x \in S \text{ and } x \notin A\}$$

Proposition 22: Let S be a set. Let A and B be subsets of S . Then $A \cap B$, $A \cup B$, and $\complement A$ are all subsets of S .

Proposition 23: Let A and B be sets. Then $A \cap B$ is a subset of A .

Proof: Let x be a member of $A \cap B$. Then $x \in A$ and $x \in B$. In particular, $x \in A$. It follows that if $x \in A \cap B$, then $x \in A$. Therefore, $A \cap B \subseteq A$.

Proposition 24: Let A and B be sets. Then A is a subset of $A \cup B$.

Proposition 25: Let A , B , and C be sets. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Proposition 26 (Commutative Property): Let S be a set and let A and B be subsets of S . Then

- (1) $A \cup B = B \cup A$ and
- (2) $A \cap B = B \cap A$.

Proposition 27 (Associative Property): Let S be a set, and let A , B , and C be subsets of S . Then

- (1) $(A \cap B) \cap C = A \cap (B \cap C)$ and
- (2) $(A \cup B) \cup C = A \cup (B \cup C)$.

Proposition 28 (Distributive Property): Let S be a set. Let A , B , and C be subsets of S . Then

- (1) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$ and
- (2) $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$.

Proposition 29: Let S be a set, and let A be a subset of S . Then

- (1) $A \cup \emptyset = A$ and
- (2) $A \cap \emptyset = \emptyset$.

Proposition 30: Let S be a set, and let A be a subset of S . Then

- (1) $A \cup S = S$ and
- (2) $A \cap S = A$.

Proposition 31: Let S be a set, and let A be a subset of S . Then

- (1) $A \cup \complement A = S$ and
- (2) $A \cap \complement A = \emptyset$.

Proposition 32 (DeMorgan's Laws): Let S be a set, and let A and B be subsets of S . Then

- (1) $\complement(A \cap B) = \complement A \cup \complement B$, and
- (2) $\complement(A \cup B) = \complement A \cap \complement B$.

Proposition 33: Let A , B , and C be sets such that $A = B$ and $B = C$. Then $A = C$.

Logic Rule 34: Let $p(x)$ be a sentence about x , and let q be a statement. Then

- (1) The statement “(there exists an x such that $p(x)$) and q ” means the same as “there exists an x such that $(p(x)$ and q)”.
- (2) The statement “(for all x , $p(x)$) or q ” means the same as “for all x , $(p(x)$ or q)” and

Let Δ and S be sets. For each δ in Δ , suppose A_δ is a subset of S . We define the following sets:

$$\bigcup_{\delta \in \Delta} A_\delta = \{x \mid x \in S \text{ and there exists a } \delta \in \Delta \text{ such that } x \in A_\delta\}$$

and

$$\bigcap_{\delta \in \Delta} A_\delta = \{x \mid x \in S, \text{ and for all } \delta \in \Delta, x \in A_\delta\}$$

Note that $\bigcup_{\delta \in \Delta} A_\delta$ and $\bigcap_{\delta \in \Delta} A_\delta$ are both subsets of S .

Proposition 35 (Generalized Distributive Property for Sets)

(Homework): Let Δ and S be sets. Let B be a subset of S . For each δ in Δ , suppose A_δ is a subset of S . Then

$$\left(\bigcup_{\delta \in \Delta} A_\delta\right) \cap B = \bigcup_{\delta \in \Delta} (A_\delta \cap B)$$

and

$$\left(\bigcap_{\delta \in \Delta} A_\delta\right) \cup B = \bigcap_{\delta \in \Delta} (A_\delta \cup B)$$

Proposition 36 (Generalized DeMorgan's Laws for Sets)

(Homework): Let Δ and S be sets. For each δ in Δ , suppose A_δ is a subset of S . Then

$$\mathbb{C}\left(\bigcap_{\delta \in \Delta} A_\delta\right) = \bigcup_{\delta \in \Delta} (\mathbb{C}A_\delta)$$

and

$$\mathbb{C}\left(\bigcup_{\delta \in \Delta} A_\delta\right) = \bigcap_{\delta \in \Delta} (\mathbb{C}A_\delta)$$

Axiom 37 (Equality Properties for Integers): Let a , b , and c be integers. Then

- *Reflexive Property:* $a = a$;
- *Symmetric Property:* If $a = b$, then $b = a$; and
- *Transitive Property:* If $a = b$ and $b = c$, then $a = c$.

Axiom 38 (Closure Property for Integers): If a and b are integers, then so are $a + b$ and ab .

Axiom 39 (Commutative Property for Integers): If a and b are integers, then $a + b = b + a$ and $ab = ba$.

Axiom 40 (Associative Property for Integers): If a and b are integers, then $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.

Axiom 41 (Distributive Property for Integers): If a , b , and c are integers, then $(a + b)c = ac + bc$.

Axiom 42 (Identity Elements for Integers): If a is an integer, then $a+0 = a$ and $A \times 1 = a$.

Axiom 43 (Additive Inverse of Integers): For each integer a , there exists an integer b such that $a + b = 0$.

The integer b in Axiom 43 is called an *additive inverse* of a .

Axiom 44 (Equation Rules for Integers): Let a , b , and c be integers such that $a = b$. Then $a + c = b + c$ and $ac = bc$.

Axiom 45 (Cancellation Law for Integers): If a , b , and c are integers such that $ac = bc$ and $c \neq 0$, then $a = b$.

Proposition 46: Let a , b , and c be integers. Then $a(b + c) = ab + ac$.

Proposition 47: If a is an integer, then $0a = 0$.

Proof: Let a be an integer. By the Identity Axiom, $0 + 0 = 0$. Thus, by the Equation Rules, $(0+0)a = 0a$. Using the Distributive Property, $0a + 0a = 0a$. Let b be an integer such that $0a + b = 0$. Then $(0a + 0a) + b = 0a + b$. Using the Associative Property, $0a + (0a + b) = 0a + b$. Thus $0a + 0 = 0$; that is, $0a = 0$.

Proposition 48: Let a , b , and c be integers. If $a + b = 0$ and $a + c = 0$, then $b = c$.

Proposition 48 tells us that each integer has a unique additive inverse. Let a be an integer. We will write $-a$ to denote the additive inverse of a . Thus $a + (-a) = 0$.

Definition 49: Let a and b be integers. We define $a - b$ by

$$a - b = a + (-b)$$

Definition 50: Let a and b be integers. We write $a < b$ if $b - a$ is a natural number. If $a < b$, we also write $b > a$. We also write $a \leq b$, or $b \geq a$, if either $a < b$ or $a = b$.

Proposition 51: Let a be an integer. Then a is a natural number if and only if $a > 0$.

Axiom 52 (Closure Property for Natural Numbers): If a and b are natural numbers, then so are $a + b$ and ab .

Axiom 53 (Trichotomy Law for Integers): For every integer a , either $a > 0$, $a = 0$, or $a < 0$.

Axiom 54: The integer 1 is a natural number. Moreover, if n is any natural number, then $1 \leq n$.

Proposition 55: The integer 0 is its own additive inverse.

Proposition 56: Let a be an integer. If $-a = a$, then $a = 0$.

Proposition 57: Let a be an integer. Then $-1a = -a$.

Proposition 58: Let a be an integer. Then $-(-a) = a$.

Proposition 59: Let a and b be integers. Then $(-a)(-b) = ab$.

Proposition 60: Let a and b be integers. If $ab = 0$, then $a = 0$ or $b = 0$.

Proof: Assume that $ab = 0$. Either $a = 0$ or $a \neq 0$. If $a = 0$, we are done! Suppose that $a \neq 0$. Then $ab = 0$ and $0 = 0a$. Thus, $ab = 0a$; that is, $ba = 0a$. By the Cancellation Law, we have $b = 0$.

Proposition 61: Let a , b , and c be integers. Suppose $a < b$ and $c > 0$. Then $ac < bc$.

Proposition 62: Let a , b , and c be integers. Suppose $a < b$ and $c < 0$. Then $ac > bc$.

Hint: Show that $-c > 0$.

A set of integers A is said to have a least element l if l is a member of A , and for all x in A , $l \leq x$.

Axiom 63 (Well-ordering Property for Natural Numbers): Every non-empty subset of the natural numbers has a least element.

Theorem 64 (First Principle of Mathematical Induction): Let S be a subset of $\mathbf{N}$ which satisfies the following two conditions:

- (1) $1 \in S$; and
- (2) If $n \in S$, then $n + 1 \in S$.

Then $S = \mathbf{N}$.

Hint: Let T be the set of natural numbers not in S . Either $T = \emptyset$ or $T \neq \emptyset$. If $T = \emptyset$, we are done!

Theorem 65 (Second Principle of Mathematical Induction): Let S be a subset of $\mathbf{N}$ which satisfies the following two conditions:

- (1) $1 \in S$; and
- (2) For each natural number n , if the set $\{k \mid k \in \mathbf{N} \text{ and } k < n\} \subseteq S$, then $n \in S$.

Then $S = \mathbf{N}$.

Theorem 66: For each natural number n , let $P(n)$ be a statement. Suppose the following two statements are true:

- (1) $P(1)$ is a true statement; and
- (2) For each natural number n , $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for all n .

Theorem 67: For each natural number n , let $P(n)$ be a statement. Suppose the following two statements are true:

- (1) $P(1)$ is a true statement; and
- (2) For each natural number n , if $P(k)$ is true for every natural number $k < n$, then $P(n)$ is also true.

Then $P(n)$ is true for all n .

Problem 68: Prove that for each natural number n ,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Problem 69: Prove that for each natural number n ,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Definition 70: Let a , b , and c be integers. We say that a *divides* b , and write $a \mid b$, if there is an integer c such that $b = ac$. If a does *not divide* b , we write $a \nmid b$.

Proposition 71: Let a , b , and c be integers. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proposition 72: Let a , b , c , m , and n be integers. If $c \mid a$ and $c \mid b$, then $c \mid (am + bn)$.

Lemma 73: Let a and b be integers such that $0 \leq a < b$. If $b \mid a$, then $a = 0$.

Theorem 74 (Division Algorithm): Let a and b be integers such that $b > 0$. Then there exist unique integers q and r such that $0 \leq r < b$ and $a = bq + r$.

Hint: You must show (1) that q and r exist, and (2) that q and r are unique. To prove part (1), let S be the set of all nonnegative integers of the form $a - bk$, where k is an integer. Show that $a \in S$ or $a - ab \in S$. Thus S is not empty. Let r be the least element of S . Continue $\dots$. To prove part (2), suppose that $a = bq + r$, $a = b'q' + r'$, and $0 \leq r' \leq r < b$. Do some algebra.

Let n be an integer. If $2 \mid n$, then we say n is *even*. If $2 \nmid n$, then we say n is *odd*.

Proposition 75: Let a and b be integers with $b > 0$. Let $a = bq + r$ such that $0 \leq r < b$. Then $b \mid a$ if, and only if, $r = 0$.

Theorem 76 (Recursion Principle): Let S be a set. For each natural number n , let d_n be a member of S . Suppose the following two conditions are satisfied:

- (1) d_1 is defined; and
- (2) for each natural number n , if d_n is defined, then d_{n+1} is also defined.

Then d_n is defined for each natural number n .

Definition 77: Let n be a natural number. We define $0!$ and $n!$ by

$$0! = 1,$$

and

$$n! = n(n-1)!$$

Definition 78: Let p be a natural number such that $p > 1$. We say that p is *prime* if the only natural numbers which divide p are 1 and p . If n is a natural number, $n > 1$, and n is not prime, then we say that p is *composite*.

Proposition 79: Every natural number greater than 1 has a prime divisor.

Hint: Let S be the set of natural numbers greater than 1 which do not have a prime divisor. We want S to be empty. Suppose S is not empty ...

Theorem 80: For each natural number n , there is a prime p such that $p \leq n$.

Theorem 80 asserts that there are infinitely many primes.

Proposition 81 (Homework): Let n be a natural number. There are consecutive integers $m, m+1, m+2, \dots, m+n-1$, none of which are prime.

Conjecture 82: Let n be an even natural number greater than 2. Then there exist primes p and q such that $n = p + q$.

Definition 83: Let a and b be integers such that $a \neq 0$ or $b \neq 0$. The greatest common divisor of a and b , $\gcd(a, b)$, is defined to be that largest natural number d such that $d \mid a$ and $d \mid b$. That is, if $d = \gcd(a, b)$, $l \mid a$, and $l \mid b$, then $l \leq d$. In addition, $\gcd(0, 0)$ is defined to be 0. Observe that $\gcd(a, b) = \gcd(b, a)$.

Problem 84: Calculate $\gcd(-48, 36)$.

Definition 85: Two integers a and b are said to be *relatively prime* if $\gcd(a, b) = 1$.

Proposition 86: Let a, b, d, m , and n be integers such that $d = \gcd(a, b)$, $a = dm$, and $b = dn$. If $d \neq 0$, then $\gcd(m, n) = 1$.

Proposition 87: Let a, b , and c be integers. Then $\gcd(a + bc, b) = \gcd(a, b)$.

Theorem 88: Let a and b be integers such that $a \neq 0$. Let $d = \gcd(a, b)$. Then there exist integers m and n such that $d = ma + nb$.

Hint: Note that $d \neq 0$. Let $S = \{xa + yb \mid x, y \in \mathbf{Z} \text{ and } ax + by > 0\}$. Show that $S \neq \emptyset$. Let l be the least member of S . Write $l = ma + nb$ where $m, n \in \mathbf{Z}$. Use the division algorithm to show that $l \mid a$. Similarly, $l \mid b$. This one may take some work.

Proposition 89: Let a, b, d , and k be integers. Assume that $d = \gcd(a, b)$, $k \mid a$, and $k \mid b$. Then $k \mid d$.

Theorem 90 (Euclid, c. 350 B.C.): Let a, b, q , and r be integers such that $0 \leq r < b$ and $a = bq + r$. Then

$$\gcd(a, b) = \gcd(a, r)$$

Problem 91 (Homework): Find $\gcd(102, 222)$ and $\gcd(20785, 44350)$.

Lemma 92: Let a, b , and c be natural numbers. If $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Lemma 93: Suppose p is prime, and $a_1, a_2, \dots, a_n$ are all natural numbers. If $p \mid a_1 a_2 \cdots a_n$, then there is a integer i such that $1 \leq i \leq n$ and $p \mid a_i$.

Hint: You will need to use Theorem 67 applied to n , the number of factors $a_1, a_2, \dots, a_n$, and Lemma 93. Show things work for $n = 1$ and $n = 2$. Assume that for $2 < k < n$, if $p \mid a_1 a_2 \cdots a_k$, then there is a integer i such that $1 \leq i \leq k$ and $p \mid a_i$. Prove that if $p \mid a_1 a_2 \cdots a_n$, then there is a integer i such that $1 \leq i \leq n$ and $p \mid a_i$.

We've got the biggie coming up next time.

Theorem 94 (Fundamental Theorem of Arithmetic): Let n be a natural number greater than 1. There exist a unique natural number k and unique primes $P_1, P_2, \dots, P_k$ such that $P_1 \leq P_2 \leq \dots \leq P_k$ and

$$n = P_1 P_2 \cdots P_k$$

We will prove this one in class.

The factorization $n = P_1 P_2 \cdots P_k$ in the Fundamental Theorem is called the *prime factorization* of n . Usually, we write the prime factorization of n in the form

$$n = \rho_1^{l_1} \rho_2^{l_2} \cdots \rho_m^{l_m}$$

where m is a natural number, $l_1, l_2, \dots, l_m$ are natural numbers, $\rho_1, \rho_2, \dots, \rho_m$ are primes, and $\rho_1 < \rho_2 < \dots < \rho_m$.

Problem 95: Write the prime factorization of 15444.

Definition 96: Let A and B be sets. We define the *Cartesian Product* of A and B , $A \times B$, to be the collection of all pairs (a, b) such that a is a member of A and b is a member of B . We call (a, b) an *ordered pair*.

Axiom 97: Let A and B be sets. Then $A \times B$ is a set.

In light of Axiom 97, we may write

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Axiom 98: Let A and B be sets. Let (a, b) and (c, d) be members of $A \times B$. Then $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Problem 99: Let $A = \{a, b, c, d, e\}$ and $B = \{a, e, i, o, u\}$. List the members of $A \times B$.

Definition 100: Let A and B be sets. A subset f of $A \times B$ is said to be a *function* (or *map*) from A into B if the following two conditions are satisfied:

- (1) For each $a \in A$, there exists $b \in B$ such that $(a, b) \in f$; and
- (2) For each $a \in A$ and for all $b_1, b_2 \in B$, if $(a, b_1) \in f$ and $(a, b_2) \in f$, then $b_1 = b_2$.

If f is a function from A into B , we write $f : A \rightarrow B$. The set A is called the *domain* of f . If $(a, b) \in f$, we write $b = f(a)$. The set

$$\{b \in B \mid \text{there exists } a \in A \text{ such that } b = f(a)\}$$

is called the *range* of f .

Problem 101: Let $A = \{a, b, c, d, e\}$ and $B = \{a, e, i, o, u\}$.

Let $f = \{(a, i), (b, i), (c, a), (d, i), (e, e)\}$.

- (1) Is f a function from A into B ?
- (2) Find the domain of f .
- (3) Find the range of f .
- (4) Find $f(d)$.

Problem 102: Let $A = \{a, b, c, d, e\}$ and $B = \{a, e, i, o, u\}$.

Let $f = \{(a, e), (b, a), (c, o), (e, a)\}$. Is f a function from A into B ? Explain your answer.

Problem 103: Let $A = \{a, b, c, d, e\}$ and $B = \{a, e, i, o, u\}$.

Let $f = \{(a, e), (b, a), (c, o), (c, a), (d, u), (e, a)\}$. Is f a function from A into B ? Explain your answer.

Problem 104: Let $A = \{a, b, c, d, e\}$ and $B = \{a, e, i, o, u\}$.

Let $f = \{(a, a), (b, b), (c, c), (i, i), (d, d), (e, e)\}$. Is f a function from A into B ? Explain your answer.

Definition 105: Let $f : A \rightarrow B$ be a function. f is defined to be *one-to-one* ($1 - 1$) if the following condition is satisfied:

For all a_1 and a_2 in A , and b in B , if $(a_1, b) \in f$ and $(a_2, b) \in f$, then $a_1 = a_2$.

Problem 106: Let $A = \{w, x, y, z\}$ and $B = \{a, e, i, o, u\}$. Give examples of two functions from A into B , one of which is $1 - 1$ and one that is not.

Definition 107: Let $f : A \rightarrow B$ be a function. f is defined to be a function from A onto B if the following condition is satisfied:

For each b in B , there is an a in A such that $(a, b) \in f$; that is, $b = f(a)$.

Problem 108: Let $A = \{u, v, w, x, y, z\}$ and $B = \{a, e, i, o, u\}$. Give examples of two functions from A into B – one of which is from A onto B , and one that is not.

Problem 109: Let $A = \{u, v, w, x, y, z\}$ and $B = \{a, e, i, o, u\}$. Is there a $1 - 1$ function from A into B ? Explain your answer.

Problem 110: Let $A = \{w, x, y, z\}$ and $B = \{a, e, i, o, u\}$. Is there a function from A onto B ? Explain your answer.

Problem 111: Let $A = \{a, b, c, d, e\}$ and $B = \{a, e, i, o, u\}$. Find a $1 - 1$ function from A onto B .

Definition 112: Let $g : A \rightarrow B$ and $f : B \rightarrow C$ be functions. We define the *composition* of f and g ($f \circ g$) to be the set

$$\{(a, c) \mid a \in A, c \in C, \text{ and there exists } b \in B \text{ such that } (a, b) \in g \text{ and } (b, c) \in f\}$$

Proposition 113: Let $g : A \rightarrow B$ and $f : B \rightarrow C$ be functions. Then $f \circ g$ is a function from A into C .

Proposition 114 (Homework): Let $g : A \rightarrow B$ and $f : B \rightarrow C$ both be $1 - 1$ functions. Then $f \circ g$ is also a $1 - 1$ function.

Proposition 115 (Homework): Let $g : A \rightarrow B$ and $f : B \rightarrow C$ be functions from A onto B and from B onto C respectively. Then $f \circ g$ is a function from A onto C .

Definition 116: Let $f : A \rightarrow B$ be a function. Define f^{-1} by

$$f^{-1} = \{(b, a) \mid (a, b) \in f\}$$

The set f^{-1} is called the *inverse* of f .

Proposition 117: Let $f : A \rightarrow B$ be a 1 – 1 function from A onto B . Then f^{-1} is a 1 – 1 function from B onto A .

Hint: You need to show that f^{-1} is a function, f^{-1} is 1 – 1, and f^{-1} is onto.

Proposition 118 (Homework): Let $f : A \rightarrow B$ be a 1 – 1 function from A onto B . Then $(f^{-1})^{-1} = f$.

Proposition 119: Let $f : A \rightarrow B$ and $g : A \rightarrow B$ be functions. Then $f = g$ if and only if for all a in A , $f(a) = g(a)$.

Proposition 120: Let $h : A \rightarrow B$, $g : B \rightarrow C$, and $f : C \rightarrow D$ be functions. Then $f \circ (g \circ h) = (f \circ g) \circ h$.

Definition 121: Let A and B be sets, and let n be natural number.

- (1) We say that $|A| \leq |B|$ if there is a 1 – 1 function from A into B .
- (2) We say that $|A| = |B|$ if $|A| \leq |B|$ and $|B| \leq |A|$.
- (3) We say that $|A| < |B|$ if $|A| \leq |B|$, but there is not a 1 – 1 function from B into A .
- (4) We say that $|A| = n$ if there is a 1 – 1 function from A onto the set

$$\{x \mid x \in \mathbf{N} \text{ and } x \leq n\}$$

- (5) We define $|\emptyset|$ by $|\emptyset| = 0$.

The notation $|A|$ is called the *cardinality* of A . Note that $|A| = |B|$ if and only if $|B| = |A|$. We say that A is a *finite set* if $|A| = n$. If A is not finite, we say that A is *infinite*.

We will not prove the next theorem.

Theorem 122 (The Schröder-Bernstein Theorem): Let A and B be sets. Then $|A| = |B|$, if and only if there is a 1 – 1 function from A onto B .

Proposition 123: Let A be a set. Then $|A| = |A|$.

Proposition 124: Let A , B , and C be sets. If $|A| \leq |B|$ and $|B| \leq |C|$, then $|A| \leq |C|$. Moreover, if $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$.

Definition 125: Let A be an infinite set. We say that A is *countable* if $|A| = |\mathbf{N}|$. If A is not countable, we say that A is *uncountable*.

Proposition 126: The set $\mathbf{Z}$ is countable.

Proposition 127: The set $\mathbf{Z} \times \mathbf{Z}$ is countable.

Proposition 128: The set of even natural numbers ($2\mathbf{N}$) is countable.

Proposition 129: Let A be a countable set. Let B be an infinite subset of A . Then B is countable.

Proof: Write $A = \{a_1, a_2, \dots, a_n, \dots\}$. Let i_1 be the least natural number such that $a_{i_1} \in B$. Let i_2 be the least natural number such that $a_{i_2} \in B \cap \mathbb{C}\{a_{i_1}\}$. Let i_3 be the least natural number such that $a_{i_3} \in B \cap \mathbb{C}\{a_{i_1}, a_{i_2}\}$. In general, let i_k be the least natural number such that $a_{i_k} \in B \cap \mathbb{C}\{a_{i_1}, a_{i_2}, \dots, a_{i_{k-1}}\}$. It follows that we may write $B = \{a_{i_1}, a_{i_2}, \dots, a_{i_k}, \dots\}$. Define $f : \mathbf{N} \rightarrow B$ by $f(k) = a_{i_k}$. Note that f is a 1-1 function from $\mathbf{N}$ onto B . Thus $|B| = |\mathbf{N}|$.

Axiom 130 (Principle of Addition): Let A and B be two sets such that $A \cap B = \emptyset$. Let m and n be natural numbers. If $|A| = m$ and $|B| = n$, then $|A \cup B| = m + n$.

Axiom 131 (Principle of Multiplication): Let A and B be sets. Let m and n be natural numbers. If $|A| = m$ and $|B| = n$, then $|A \times B| = mn$.

Problem 132 (Homework): Let $A = \{a, b, c, d\}$ and $B = \{e, f, g, h, i\}$. Find $|A \cup B|$.

Problem 133 (Homework): Let $A = \{a, b, c, d\}$ and $B = \{b, d, e, f\}$. Find $|A \cup B|$.

Problem 134 (Homework): Let $A = \{a, b, c, d\}$ and $B = \{e, f, g, h, i\}$. Find $|A \times B|$.

Problem 135 (Homework): Let $A = \{a, b, c, d\}$ and $B = \{b, d, e, f\}$. Find $|A \times B|$.

Problem 136 (Homework): Let A be a set. Find $|A \times \emptyset|$.

Definition 137: Let A be a set. A subset R of $A \times A$ is said to be an *equivalence relation* on A if the following conditions are satisfied:

- *Reflexive Property:* For each $a \in A$, $(a, a) \in R$;
- *Symmetric Property:* For all $a, b \in A$, if $(a, b) \in R$, then $(b, a) \in R$; and
- *Transitive Property:* For all $a, b, c \in A$, if $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

Problem 138: Let $A = \{a, b, c, d\}$. Let

$$R = \{(a, a), (a, c), (a, d), (c, a), (c, c), (c, d), (d, a), (d, c), (d, d)\}$$

Is R an equivalence relation on A ?

Problem 139: Let $A = \{a, b, c, d\}$. Give an example of a subset of $A \times A$ which satisfies the reflexive and symmetric properties, but is not an equivalence relation.

Problem 140: Let $A = \{a, b, c, d\}$. Give an example of a subset of $A \times A$ which satisfies the symmetric and transitive properties, but is not an equivalence relation.

Problem 141: Let $A = \{a, b, c, d\}$. Give an example of a subset of $A \times A$ which satisfies the reflexive and transitive properties, but is not an equivalence relation.

Problem 142: Let A be a set. Let $R = A \times B$. Is R an equivalence relation on A ?